

FIELD CHECKLIST / FREE DOWNLOAD

From Equipment Purchase to Long-Term O&M

A practical gas-engine project checklist

Six evidence gates for buyers, owners, investors and lenders - from the first equipment decision through commissioning, operating data, maintenance and spare-parts readiness.

01

DEFINE

02

VERIFY

03

TEST

04

CONTROL

05

BASELINE

06

MAINTAIN

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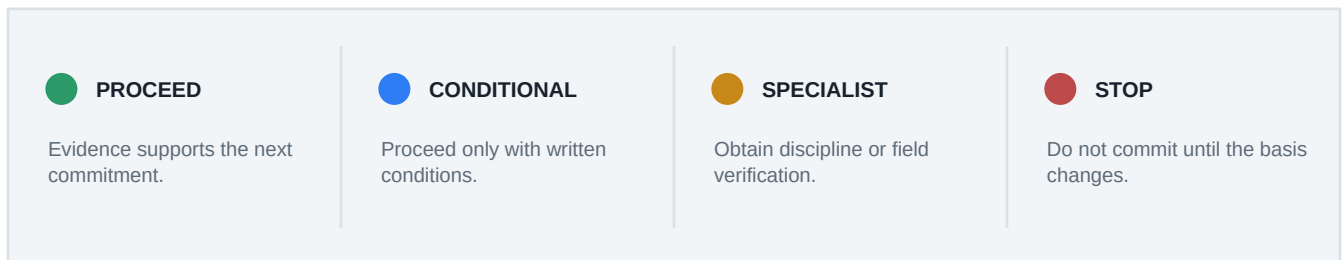
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STAGE-GATED DECISION METHOD

Move on evidence, not calendar milestones

At each gate, define the next commercial commitment, the minimum evidence required, the owner of every open item and the consequence of proceeding without it.



The six connected gates

01 Define the project decision

Decision, boundary, risk owner and evidence threshold

02 Verify equipment before purchase

Identity, hours, preservation, condition and exclusions

03 Test project and financial assumptions

Fuel, site, configuration, O&M and sensitivity basis

04 Control delivery and acceptance

Receipt, storage, installation, commissioning and acceptance

05 Establish the operating baseline

Runtime, production, meters, events and configuration history

06 Maintain the asset deliberately

Maintenance framework, interventions, spares and plan updates

PROJECT LIFECYCLE

Evidence gates 1-2

GATE 01

Define the project decision

- ☐ State the required electrical output, operating mode and decision timeline.
- ☐ Identify who bears fuel, grid, performance, logistics and maintenance risk.
- ☐ Separate verified inputs from assumptions and commercial aspirations.
- ☐ Define the evidence required before equipment commitment.

DECISION / COMMITMENT	EVIDENCE GAP	OWNER AND DUE DATE

GATE 02

Verify equipment before purchase

- ☐ Review serial-specific nameplates, configuration and document consistency.
- ☐ Check operating-hour evidence and whether controller information can be corroborated.
- ☐ Assess preservation, corrosion, visible condition and inspection access.
- ☐ Use borescope or other physical inspection when internal condition affects value.
- ☐ Record exclusions and unresolved questions in the purchase decision.

DECISION / COMMITMENT	EVIDENCE GAP	OWNER AND DUE DATE

PROJECT LIFECYCLE

Evidence gates 3-4

GATE 03

Test project and financial assumptions

- ☐ Map site fuel quality, pressure, conditioning and metering basis.
- ☐ Check ambient, cooling, ventilation, exhaust, emissions and acoustic constraints.
- ☐ Reconcile engine configuration with the electrical and balance-of-plant concept.
- ☐ Build maintenance, overhaul, spares and downtime assumptions into the model.
- ☐ Run sensitivities instead of relying on one fuel, output or availability case.

DECISION / COMMITMENT	EVIDENCE GAP	OWNER AND DUE DATE

GATE 04

Control delivery and acceptance

- ☐ Inspect equipment identity and condition on receipt.
- ☐ Track discrepancies, preservation and storage.
- ☐ Confirm installation prerequisites and current documentation.
- ☐ Define commissioning hold points, test records and acceptance criteria.
- ☐ Do not confuse arrival, installation, first start or synchronization with final acceptance.

DECISION / COMMITMENT	EVIDENCE GAP	OWNER AND DUE DATE

PROJECT LIFECYCLE

Evidence gates 5-6

GATE 05

Establish the operating baseline

- ☐ Retain unit-specific runtime, production, alarm and event records.
- ☐ Record meter boundaries, units, reference conditions and data quality.
- ☐ Classify planned, technical, fuel, grid, weather and unknown downtime.
- ☐ Keep benchmark calculations separate from measured performance.
- ☐ Preserve configuration and control changes for later troubleshooting.

DECISION / COMMITMENT	EVIDENCE GAP	OWNER AND DUE DATE

GATE 06

Maintain the asset deliberately

- ☐ Use a maintenance framework aligned with actual configuration and duty.
- ☐ Plan routine work and major interventions without inventing universal intervals.
- ☐ Build a risk-based spare-parts strategy and preserve inventory.
- ☐ Review recurring events and unit-to-unit differences.
- ☐ Update the financial and operating plan when evidence changes.

DECISION / COMMITMENT	EVIDENCE GAP	OWNER AND DUE DATE

LIFECYCLE RISK SCREEN

Stop weak evidence before it becomes sunk cost

A red flag is not automatically a rejection. It is a reason to identify the unsupported assumption, quantify its consequence and assign the next verification action.

- ☐ A fleet is sold as homogeneous but serial or nameplate evidence differs.
- ☐ Operating hours rely on one unsupported controller reading.
- ☐ Efficiency is calculated from inconsistent gas totals.
- ☐ An availability percentage has no defined denominator or exclusions.
- ☐ Commissioning evidence consists only of start-up screenshots.
- ☐ Quoted spare parts are treated as delivered and usable.
- ☐ Site constraints are deferred until after equipment purchase.
- ☐ Unknown downtime is hidden rather than investigated.

EVIDENCE BOUNDARY

The checklist is a decision aid. It does not replace project-specific engineering, legal, financial or safety review. Keep measured records, OEM information, contract values and K4-Technology analysis separate.

NEXT VERIFICATION ACTION

CHOOSE THE SMALLEST USEFUL SCOPE

Match the review to the decision

Remote review is suitable for bounded document, data and model questions. Installed condition, identity, commissioning response, meter boundaries and site constraints may require physical inspection or specialist field support.

QUICK REMOTE REVIEW

USD 450

Initial red-flag screen for a bounded evidence package or equipment offer.

FULL REMOTE REVIEW

USD 950

Broader reconciliation of documents, serial evidence, operating data and written conclusions.

EXPRESS MODEL REVIEW

FROM USD 450

Targeted review of fuel, output, availability, maintenance or lifecycle assumptions.

FULL MODEL REVIEW

FROM USD 1,500

Deeper review of the model basis, technical inputs, sensitivities and evidence gaps.

FEASIBILITY SCREENING

FROM USD 1,250

Early project screen connecting fuel, equipment, site, output, O&M and economics.

FULL FEASIBILITY STUDY

FROM USD 4,000

Structured technical-economic basis for a defined project decision and evidence set.

FIELD SUPPORT / QUOTED SCOPE

Use physical inspection or specialist coordination when reliable records cannot establish installed condition, internal condition, identity, commissioning response, meter boundary or site constraints.

BEFORE THE NEXT COMMITMENT

Final decision checklist

- ☐ What decision is being made now?
- ☐ Which evidence is verified, contracted, methodological or commercial only?
- ☐ Which uncertainty can be closed remotely?
- ☐ Which uncertainty requires physical inspection?
- ☐ What is the consequence of proceeding without the missing evidence?
- ☐ Who will own the open item and by when?
- ☐ Does the next commercial commitment depend on closing it?

DECISION		
PROCEED / CONDITIONAL / SPECIALIST / STOP	OWNER	DUE DATE

NEED AN EVIDENCE-BASED REVIEW?

Send the evidence pack and the decision you need to make.

Remote technical review, feasibility assessment, model review and owner-side specialist coordination are scoped to the decision and available evidence.

[REQUEST A TECHNICAL REVIEW](#)

Independence statement

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